

Institute of Actuaries of India

Subject SP4 – Pensions & Other Benefits

November 2020 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:**i) Consideration in designing DB Scheme**

- Type of Defined Benefit Scheme
- e.g. CARE etc.
- Level & generosity of benefits
- This is primarily accomplished through the accrual rate which is the amount of benefits built up for each year of service
- Eligibility
- e.g. an individual may be required to reach a minimum age before becoming eligible to accrue or receive certain benefits etc.
- This may help to reduce the costs of provision and in some cases may not significantly affect the benefit needs of an individual.
- Vesting periods may mean that short serving employees aren't eligible for benefits which will reduce costs etc
- Dependent's provision
- Risk Benefits
- E.g. benefits on death in service or ill-health.
- Death benefits are commonly in the form of a lump sum to cover immediate liabilities such as funeral costs and perhaps loan repayments[½]
- Together with a pension to provide continued financial protection for dependents.
- Integration with other benefits
- For all types of non-State benefit, the provider may wish to target a total level of benefit by allowing for State benefits in setting their own level of benefit provision.
- As individuals are likely to join and leave at different ages, it is unlikely that this will be possible in an accurate manner.
- Pension Increases / Revaluation in deferment
- Consideration should be given to whether a benefit will continue to meet the needs of the beneficiary in future years.
- It may therefore be appropriate to increase the amount of the benefit from time to time on a regular basis, e.g. annually.
- This could be at a fixed rate, in line with some form of inflation index, on a discretionary basis, or a combination of these.
- Benefits on Leaving Employment
- The State and employers may choose to differentiate between those who leave employment of their own choice and those who are forced to leave due to company restructuring or a general economic downturn.
- In relation to the former group, it may be felt that there is no obligation to provide any benefit.
- Alternatively, it may be viewed that where any benefit promise has been made there is a need to maintain equity between any employees who leave employment and those who remain.
Target retirement age (maybe for workforce planning)
- Definition of pensionable salary
- Link to salary and service
- Contributions (whether to get engagement/save costs by asking employees to contribute and how much)
- Options and guarantees
- Target pension at retirement
- Regulation (need to ensure benefit structure is compliant)
- Integration with other employer benefits i.e. fitting into a flex scheme
- Tax efficiency – design it to be as tax efficient as possible

[0.5 each, Max 7]

ii) Minimum period of employment

Advantages

- It produces less administration for short stayers who never become eligible for membership. [0.5]
 It will save the employer money, as all members build up less service in the scheme. [0.5]
 It can help the employer to meet their aims, *eg* target resources to long-term staff. [0.5]

Disadvantages

- It may be more difficult to persuade someone to join your company if they have to wait to join the pension scheme which is the primary objective of the employer as stated in the question hence employer objective may be defeated to attract the staff. [1]
 When employees are unable to join the pension scheme, they may be persuaded to make their own individual arrangements and thus be less likely to join the company scheme when they are eligible. This may not serve their best interests and defeats the employer objective to retain the staff. [0.5]
 Those in greatest need of the death-in-service benefit, *eg* young people with family responsibilities, may not be covered. [0.5]
 This problem could be resolved by providing life assurance benefits only in the waiting period. [0.5]
 Where the scheme requires employees to contribute, they are less likely to join after a waiting period since they will have become used to a certain level of net income. [0.5]
 It is difficult to achieve a waiting period that is equitable between different groups, *eg* should it be the same length for part timers? [0.5]

[Max 4]

iii) Reducing risks of the Scheme

- Set up DB scheme as a CARE scheme or cash balance etc. to mitigate salary risk
- Address operational risks e.g. outsource admin
- Have some benefits discretionary
- Require consent for options
- Risk reduction exercises e.g. PIE etc.
- The scheme could be changed to be a defined contribution scheme.
- In this case the experience risks, particularly of investment and (pre-retirement) mortality lie with the individual.
- However, this may be very unpopular with employees who value the security of a DB scheme and are concerned about the risks they are taking on in a DC scheme
- Hybrid Scheme – Alternatively a scheme design that contains elements of risk sharing with employees may be feasible.
- For example this could mean a scheme that is DB to a certain salary level and DC after that
- Or one which has a set cap on employer costs with employees paying the balance.
- Contributions – Increasing the level of employee contributions will reduce the cost for the employer
- In particular some sort of “cost sharing” mechanism will reduce the risk of large future contributions [½]
- For example the employee contributions could be set to automatically increase when the employer contributions increase
- Financing – The company could finance benefits more prudently
- This would reduce the risk of unexpected future requirements for funding
- It may also reduce the level of costs if the investments generate real returns
- Investment Strategy - Investing in lower risk assets will reduce the risk of investment under performance but at the cost of lower expected returns.
- Some risks may be mitigated through the investment strategy; for example purchasing Longevity Bonds.
- NRA could be adjusted in line with trends in the general population.
- The company can also look to match their assets to the liabilities, e.g. by investing in assets that are expected to increase at the same rate as pension payments.

- Insurance – The company could insure some benefits such as death in service and ill-health to remove the volatility of costs associated with this.
- The company could also purchase annuities from insurance companies to cover the pension payments although this is likely to be expensive

[0.5 each, Max 8]

iv) Cash balance schemes

A percentage of salary is set aside each year for each member. Each year's accrual is indexed up to retirement at a guaranteed minimum rate, for example in line with price inflation or investment returns.

The member's fund is available at retirement to secure retirement benefits (for example by purchasing an annuity with an insurance company). The scheme could be viewed as similar to a final salary scheme, except the targeted benefits are in the form of a lump sum instead of a pension.

[3]

v) Options to the members

- Commutation
- Early Retirement
- Exchange of dependents pension / Allocation
- Late retirement
- CETVs
- Changing level / form of benefits
- E.g. Options on pension increases (e.g. akin to PIE)

[0.5 each, Max 2]

vi) Risks of introduction of options to the members

- Uncertainty of cost
- Matching difficulties e.g. cash flows, liquidity requirements etc.
- Workforce planning difficulties
- Risk of selection against the scheme e.g. if members with low life expectancy commute high levels of benefits
- Selection risk, may only be purchased by those who expect to gain some value from it Liquidity risk as money may be required immediately
- Longevity risk e.g. dependents live longer than expected
- Inflation risk e.g. PIE to inflation linked pension
- Risk that options factors, e.g. early retirement factors, are not cost neutral so overall benefits paid out are higher
- May encourage workers to leave early which could have workforce planning implications [½]
- May be additional administrative requirements, particularly if the additional benefits are of a different form to the scheme benefits.
- Reputational risk if option terms are deemed to be unfair
- Overall strain if terms are deliberately set to be generous

[0.5 each, Max 4]

vii) Monitoring the strength of the sponsor covenant

- Review any financial information available, for example the sponsor's published accounts.
- Monitor the level of any risk-based levies, for example risk-based premiums to a central discontinuance fund.
- Maintain regular contact with the finance director and board of the sponsoring company.
- Require that the sponsor informs the trustees of any actions it takes or is intending to take that could materially impact on the strength of the covenant.
- Ensure the strength of the covenant is regularly monitored by qualified professionals.

Actions for trustees

- Change the investment strategy to bonds to increase value of liabilities.
- Invest in assets that pay out on sponsor default, *eg* derivatives including credit default swaps.
- Consider alternatives to cash payments by sponsor, for example a charge on the sponsor's fixed assets.
- Require ratchets to be in place, so if the financial position of the sponsor improves then the contributions to the scheme increase to reflect this improvement.
- Require contingent contributions, so the deficit needs to be made up more quickly if the scheme's financial position worsens.

[0.5 each, Max 5]

[33 Marks]**Solution 2:****i) Criteria for choosing funding strategy**

Stability – stability of contribution rate in normal course of events.

Security – the ability to meet the benefit expectations in all events.

Durability – similar to stability; it relates to the stability of contribution rate when there is a major event, *eg* closure, lots of leavers, lots of new entrants.

Realism – a realistic measure on a year to year basis of the long-term costs of the benefits.

Liquidity – availability of sufficient funds to pay the benefits as they are due.

Flexibility – the ability of the sponsor to choose the timing of contributions.

Opportunity cost – comparative returns available on money invested in the benefit scheme and money invested elsewhere.

[0.5 each , Max 3]

ii) Considerations in determining an appropriate investment strategy*Liability profile*

The average term of the liabilities will be very long when the scheme begins. The term of appropriate assets will reduce over time.

The nature of the benefits is also important in assessing the suitability of the assets.

Early on, the liabilities will mainly be linked to earnings. As the scheme matures, more benefits may be linked to price inflation or may have fixed increases, depending on the nature of the benefits provided.

Fund size

Early on there will be limited funds available and so some investment classes may not be suitable because of their size, *eg* direct property investment. In the longer term, new classes may become available as the scheme grows.

Level of risk / security

The views of the trustees and the sponsor should be sought so as to ascertain their attitude to risk.

The strength of the sponsor covenant should also be assessed.

A more aggressive investment strategy could be adopted to try to increase returns and hence reduce costs ...

... the downside is that there is a risk of a significant deficit if experience turns out to be worse than expected, resulting in increased costs and a volatile contribution rate.

The scheme could minimise the probability of deficit by holding assets that closely match the liabilities. However, the cost of doing so may be unacceptably high in terms of lower investment returns ...

... furthermore, surpluses are less likely to arise and hence members are less likely to be granted benefit improvements.

The consequences of poor investment performance are less significant for an immature scheme ...

... it may therefore wish to follow a policy of deliberately mismatching, in order to benefit from higher expected returns from assets that do not closely match the liabilities.

The scheme may consider investing a greater proportion of the scheme in equities, which historically have outperformed government bonds over the long term, but have extra risks associated with them, including default and volatility. (Questions also need to be asked about future returns on equities versus other investments.)

Funding level

The healthier the funding position, the greater the level of investment freedom, increasing the scope to mismatch.

Liquidity

For an immature scheme, benefit outgo is low and therefore can be paid out of contribution income. [½]

Liquidity will therefore not be a major issue and the scheme will have less need to invest in marketable and high yielding assets. (Although assets with a stable market value may be desirable assuming market-related valuations are carried out).

Consideration should be given to any protection benefits offered, for example a lump sum payment on death in service could lead to liquidity risk for an immature scheme with only a small fund, which may lead to insurance being used.

Diversification

A diversified portfolio of assets reduces risk. The optimum position between diversification and minimising investment costs needs to be sought.

Restrictions / legislation

The State may restrict the types of investments held. For example, it may restrict the level of self-investment or set levels for minimum / maximum holdings of different types of investment. (The way statutory tests are carried out may also encourage certain investments, eg holding assets with a stable market value.)

Tax position

The tax implications of holding certain investments should be considered. The return net of tax is the key issue.

Investment expenses

The expenses associated with investment should also be considered. The return net of expenses is the key issue.

Ethical / moral issues

The trustees and/or sponsor may choose to avoid certain investments from an ethical / moral viewpoint.

Competitor schemes / benchmarking

The scheme is unlikely to want to risk poor performance compared to its peers, and may want to specify that the managers should not deviate significantly from the investment strategy followed by competitors' schemes (although this should not be a primary consideration).

[0.5 each, Max 9]

iii) Fixed-interest securities may be held because they are a good match for liabilities, for example they may be a good match for:

- actives, if part of the pension benefits can be taken as a fixed amount of cash at retirement

- current pensioners, if pensions are level or have fixed increases
 - deferred pensioners, if pensions have fixed increases between the date of leaving and retirement.
- The securities are issued in the domestic currency and therefore a good currency match for the liabilities. Fixed-interest securities have a high running yield, as communicated by the Finance Director, and therefore may be held if the scheme needs a higher income from its assets to match outgo (*eg* if the scheme has negative cashflow which may be the case if the scheme is quite mature).
- Fixed-interest securities may be held for diversification from other asset classes.
- Fixed-interest securities may be issued by governments or companies, the two types may be held for increased diversification.
- Fixed-interest securities (in particular government securities) may be held as it is felt they are a very secure investment (*ie* low risk of default).
- Fixed-interest corporate securities may be held as they offer a higher expected return than government securities.
- Fixed-interest securities may be held if the scheme has a policy of buying-out benefits that are level or have fixed increases through an insurance company at retirement (since annuities are backed by securities and therefore move in line with bond prices).
- A high proportion of fixed-interest securities is likely to be held by a scheme that will shortly purchase immediate and deferred annuities from an insurer.
- At certain times it may be felt that fixed-interest securities are cheap (*ie* offer attractive returns), relative to other investments.

[0.5 each, Max 5]

[17 Marks]**Solution 3:**

i)

Make a lump sum contribution

- May be required if funding level is very low
- In particular, a deficit on a discontinuance basis

Flat / level contribution

- May be appropriate if there is a discontinuance deficit that requires removal over a short period, but not immediately
- Useful for schemes where salary or headcount is volatile, e.g. small scheme or one that is reducing in size

Contributions linked to inflation

- Same circumstances as above

Additional % of salary for future working lifetime of members

- Suitable when there is no pressure on for rapid removal of deficit, e.g. no discontinuance deficit
- Consistent with standard contribution rate which is usually expressed as a % of salary

Additional % of salary for a temporary time period

- May be more appropriate than above approach if there is pressure from members and / or trustees to remove deficit more quickly

Combination of any of the above

- For example, correct part of the deficit with a lump sum payment and remove remaining deficit gradually through regular payments

Do nothing

- May be most suitable when deficit is small or no discontinuance issue
- Can be justified that experience fluctuations may mean that small surpluses and deficits are to be expected

Don't grant discretionary increases

- May be a suitable way to prevent situation from getting worse

Reduce future benefits

- May not correct the deficit
- However, prevents situation from getting worse

Wind-up scheme

- Could be used as a last resort

Altering the valuation basis or method

- Only appropriate if previous basis / method was unduly cautious. No point in "hiding" a deficit through a weak valuation basis / method

[0.5 each, Max 8]

ii)

- The immediate impact will depend on whether members retain their salary linkage on their accrued benefits.
- If members retain salary link, there is minimal impact of liabilities in the short term.
- The scheme is still likely to have a combination of liabilities – salary linked for active members immediately before closure and inflation linked / linked to fixed increases for deferred pensioners and current pensioners.
- In such circumstances, it is appropriate to back actives' liabilities with equities or index linked bonds to provide broad match for salary growth.
- Deferred pensioners and pensioners liabilities are matched with index-linked or fixed income bonds (depending on the nature of pension increase granted).
- If members do not retain salary link, then active members liabilities will be similar to deferred pensioners liabilities.
-and therefore, best matched with index-linked or fixed income bonds (depending on the nature of pension increase granted for deferred pensioners).
- Liquidity could be an immediate concern as there may be no future contributions. Bigger issue is transfer values or lump sums are offered.
- If several members take out lump sums, then remaining liabilities may be substantially reduced.
- This may mean the scheme is more risk averse, needs to ensure assets more diversified and need to move to a more matched position.
- The investment strategy may need to change to move to increased proportion of liquid, marketable assets with stable market values.
- Over time, the scheme will mature. This means:

- Gradual increase to pensioners which will need to be matched by increased holding of bonds that match pension increases in payment.
- Term of liabilities will reduce, suggesting less risk being taken and holding of shorter-term assets.
- Higher level of benefit outgo, leading to a need for more liquidity and assets with higher running yield.
- An ALM study could be conducted to determine the optimal strategy.
- As the scheme matures even more, it may be more appropriate to secure the liabilities with an insurance company, through the purchase of deferred and immediate annuities.

[0.5 each, Max 9]

iii)

Key considerations are:

Cost

- This is a key consideration
- For example, if total contribution rate is set to target same level of benefit as that from the current DB scheme, then requiring members to contribute will reduce employer's cost
- Consider tax concessions on the employee's and employer's contribution – it may be tax efficient for members to contribute
- However, there may be limits on total employee contributions that qualify for tax relief. A contributory scheme may reduce ability of members to make contributions to other pension arrangements.

Attraction and retention

- Key objective for employer.
- Non-contributory scheme may be more attractive and will encourage membership.
- However, if members contribute, they may appreciate benefits more.

Administration

- Cost and ease of administration may be important to employer.
- Administration is likely to be cheaper and easier for non-contributory scheme.

External factors

- Consider approach taken by competitors.
- Consider regulation, legislation, or professional guidance that may apply.

Following factors to consider when setting level of contributions:

- Hard to set contribution rate at higher level than current scheme, as any increase may lead to employee / union reactions.
- Any increase will need to be justified as may only be acceptable if projected benefits under DC scheme likely to be higher than current DB scheme.
- A lower contribution rate may be more appropriate if DC scheme likely to provide lower benefits as compared to DB scheme.
- However, for a DC scheme, members clearly see the value of the contributions they make, so may be possible to justify without much dissatisfaction.
- Any changes from current rate may be adjusted through other areas of remuneration.
- Employer could match member contributions up to a limit.

[0.5 each, Max 8]

iv)

Assumptions required:

- Future salary growth
- If choice of contribution rates, then future contribution rate
- Allowance for career breaks, if any
- Investment return of asset classes expected to be held
- Expenses, if met from the fund
- No pre-retirement decrement assumptions will be used
- Post-retirement investment return, if pension paid from fund
- Post-retirement mortality
- Annuity rate if pension secured from insurance company

[0.5 each, Max 3]

v)

Investment considerations

- Profile of the liabilities – there is a requirement that the investments offered are a suitable match for the liabilities in terms of nature, term, currency and certainty. [0.5]
- For active members, real assets provide a good broad match for salary growth, e.g. equity investments. Closer to retirement, the investment options must enable members to match the movements in annuity rates, e.g. bond investments. [1]
- Sophistication of members – consider the ability of members to make suitable investment decisions. If the workforce has limited financial awareness it may be sensible to offer very limited options, and a “default” option. [1]
- Legislation – the investment options offered must take account of any restrictions, e.g. overseas investment or a requirement to offer a default fund. [1]
- Limitations on holdings – to achieve diversification, and reduce mismatching, the scheme may limit minimum and maximum holdings of different investment types; e.g. individual can choose a maximum holding of 50% in equity investments. [1]
- Investment costs – consider the investment costs of offering different investment options. The more options offered the higher the cost. Also, some investment vehicles may incur high expenses. [1]
- Lifestyle switching – consider introducing an automatic gradual switch of the member’s investments over a period (say 5 years) to achieve a matched position at retirement. If the retirement benefits are a mixture of a lump-sum payment and pension then a mixture of cash and fixed-interest or index-linked bonds would be a suitable match. [1]
- Size of the scheme – a new scheme will be limited in the early years from investing directly in certain asset categories. As the scheme ages further options will be possible. [1]
- Frequency of switching – limits may be imposed on how often members can amend their investment choice to reduce administration costs. [0.5]
- Competition – consider the investment options offered by competitor schemes in the industry. [0.5]
- Ease of valuation – the ease with which different investments can be valued (to show members the value of their account) should be considered. [0.5]

[Max 8]

[36 Marks]

Solution 4:

i)

Investment strategy

- The trustees' primary concern is the security of benefits, i.e. ensuring that the benefits promised are actually paid. [1]
- The employer's primary concern is to provide benefits at an acceptable cost. [1]
- In theory, therefore, the employer may prefer a riskier investment strategy, which may lead to a higher probability of insolvency, but should in the long-term lead to a better return and hence a lower cost. [1]
- However, the employer is also concerned that its contribution rate is fairly stable and predictable. [1]

How ALM can help in determining investment strategy

- ALM can assess different asset allocation strategies by simulating thousands of times the range of possible outcomes after adopting the various different strategies ... [1]
- ... i.e. many simulations can indicate which strategy gives the best expected return for a specified level of risk. [0.5]
- Often the model is kept reasonably simple, i.e. splits assets only between "equity types" and "bond types" or between real and fixed. [0.5]
- The objectives of an investment strategy need to be set. These objectives must be clear and measurable. [½]
- For example, to maximise the investment return subject to the probability of the funding level falling below 90% being less than 1% (if contributions are paid at the standard contribution rate). [0.5]
- More generally, ALM can be used to test asset allocation strategies allowing trustees to choose their preferred trade-off between risk and reward. [1]
- The results will depend on the assumptions made, thus they should be tested for robustness by carrying out sensitivity analysis. [1]
- ALM can be used to assist in the long-term asset allocation strategy; but cannot be used to suggest short-term tactical deviations from this strategy. [1]

[Max 10]

ii)

- The asset liability modelling (ALM) exercise is a cashflow projection. A cashflow model of assets and liabilities will require more and different assumptions compared to the traditional deterministic valuation that uses the present value of assets and liabilities. [1]
- In particular assumptions are needed for items which may be ignored in the funding valuation as they are financially neutral, despite having different timing of payments. [1]
- For example, the payment of lump sum on retirement, early retirements, transfers out/in etc. may be financially neutral, but can have a significant effect on the cashflow of a pension scheme. [0.5]
- Also, the ALM exercise requires further refinements:
 - usually stochastic elements to model investment behaviour [0.5]
 - an explicit new entrant assumption [0.5]
 - a link between liability values and the random parameters in the asset model. [0.5]

[Max 4]

[14 Marks]
